

Compare response_time between the placebo and drug arms and tell me whether the difference is statistically significant.

已停止思考 >

I compared response_time between the placebo and drug arms using a two-sample Welch's t-test (which does not assume equal variances).

Summary statistics

Group	n	Mean response_time	SD
Placebo	30	2.56	2.27
Drug	30	4.10	5.22

Hypothesis test

- Null hypothesis (H₀): Mean response time is the same in both groups.
- Alternative hypothesis (H₁): Mean response time differs between groups.

Test result:

- t = 1.48
- p = 0.147

Conclusion

The drug arm had a higher average response time than the placebo arm (4.10 vs 2.56), but the difference was **not statistically significant** at the conventional 0.05 level because the p-value (0.147) is greater than 0.05.

So, based on this dataset, there is insufficient evidence to conclude that the drug changes response time relative to placebo.

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